



Map, Filter, and Reduce Problem Bank

Legend

★ = Core Placement Problem (Mandatory)

These problems should be completed by every student to gain confidence in functional-style Python programming.

Section 1: Lambda Function Basics

★ Lambda Function to Add Two Numbers

★ Lambda Function to Find Square of a Number

Lambda Function to Check Even or Odd

★ Lambda Function with Multiple Arguments

Lambda Function inside Another Function

Lambda vs Regular Function Comparison

Section 2: map() Function

★ Square Each Element of a List using map()

★ Convert List of Strings to Uppercase using map()

★ Add Two Lists Element-Wise using map()

Convert List of Integers to Strings using map()

★ Map a Lambda to Calculate Cube of Numbers

Map Function with a Named (Non-Lambda) Function

Celsius to Fahrenheit Conversion using map()

Section 3: filter() Function

★ Filter Even Numbers from a List

★ Filter Odd Numbers from a List

★ Filter Positive Numbers from a List

Filter Strings Longer than N Characters

★ Filter Prime Numbers using filter() and Lambda

Filter Palindromes from a List of Strings

Filter None Values from a List

Section 4: reduce() Function

★ Sum of List Elements using reduce()

★ Product of List Elements using reduce()

Find Maximum Element using reduce()



★ Find Minimum Element using reduce()

Concatenate List of Strings using reduce()

Find GCD of a List of Numbers using reduce()

Factorial of a Number using reduce()

Section 5: Combining map, filter, and reduce

★ Square Only Even Numbers using map() and filter()

★ Sum of Squares of Even Numbers using map(), filter(), reduce()

Filter then Map a List of Dictionaries (e.g. Student Records)

★ Find Sum of Lengths of Words Longer than 3 Letters

Filter and Sort Combined Pipeline

Chain map, filter, and reduce to Process Sales Data

Section 6: Real World Functional Programming Applications

★ Apply Discount to a List of Prices using map()

★ Filter Eligible Students Based on Marks using filter()

Calculate Total Revenue using reduce()

★ Extract Specific Field from List of Dictionaries using map()

Validate a List of Emails using filter()

★ Functional Pipeline for Employee Bonus Calculation